



Persulfidation of transcription factor MYB10 inhibits anthocyanin synthesis in red-skinned pear

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Abstract

Hydrogen sulfide (H₂S) is a gaseous signaling molecule that delays color change during fruit ripening. Whether H₂S affects anthocyanin biosynthesis in red-skinned pears (*Pyrus* L.) remains unclear. Here, we found that H₂S substantially inhibits anthocyanin accumulation in red-skinned pears and the expression of several genes encoding transcription factors is affected in response to H₂S signaling. For example, *PyMYB10* and *PyMYB73* were down-regulated, whereas *PyMYB114* and *PyMYB6* were up-regulated. Bioinformatics analysis showed that *PyMYB73* and *PyMYB6*, each containing an EAR motif, may negatively regulate anthocyanin accumulation. Transient expression analysis showed that *PyMYB73* substantially promotes anthocyanin biosynthesis by co-transforming with *PyMYB10/PyMYB114 + PybHLH3*; however, *PyMYB6* inhibited anthocyanin biosynthesis in strawberry (*Fragaria vesca*) receptacles and pear fruits, and *PyMYB73* interacted with *PyMYB10* and *PyMYB6* but not *PyMYB114* or *PybHLH3*. Further investigation showed that Cys194 and Cys218 of *PyMYB10* were modified by persulfidation and that *PyMYB10^{Cys218Ala}* substantially increased anthocyanin accumulation by a transient transformation system. Co-transformation of *PyMYB10^{Cys218Ala} + PyMYB73/PyMYB6* also promoted anthocyanin accumulation in pear fruits. Yeast two-hybrid assays showed that the mutation of *PyMYB10* did not affect the interaction between *PyMYB10* and *PyMYB73*, but it inhibited interaction with *PyMYB6*. Moreover, H₂S weakened the interaction between *PyMYB10* and *PyMYB73* but enhanced the interaction with *PyMYB6*. Thus, we provided a model in which *PyMYB10* undergoes persulfidation at Cys218, enhancing the interaction with *PyMYB6* and reducing the interaction with *PyMYB73*. These subsequently results in lower expression of the anthocyanin biosynthesis-related genes *Pyrus dihydroflavonol 4-reductase* (*PyDFR*), *Pyrus anthocyanidin synthase* (*PyANS*), *Pyrus UDP-glucose: flavonoid 3-glucosyl transferase* (*PyUGT*) and *Pyrus glutathione S-transferase* (*PyGST*), thereby inhibiting anthocyanin accumulation in red-skinned pears. Our findings provided a molecular mechanism for H₂S-mediated anthocyanin biosynthesis in red-skinned pears.

Introduction

Pear (*Pyrus* L.) is an important economic fruit in temperate regions. Recently, red-skinned pear has been widely accepted by consumers around the world because of its bright color and abundant nutritional value (Tao et al. 2018). The red color of pear skin is due to anthocyanin accumulation (Zhang et al. 2020), which is largely synthesized in plants and provides various colors and protection due to components

and complex environments for cell activities. Besides, anthocyanins have remarkable antioxidant activity and underlying benefits to human health, such as decreasing the risk of cancer, coronary atherosclerosis, and inflammation (Butelli et al. 2008; Sun et al. 2014).

Anthocyanins are final products of the flavonoid pathway. Anthocyanin biosynthesis is mediated by a sequence of enzymes, incorporating cinnamate 4-hydroxylase (C4H),

chalcone synthase (CHS), chalcone isomerase (CHI), flavonoid 3-hydroxylase (F3'H), dihydroflavonol4-reductase (DFR), anthocyanidin synthase (ANS), UDP-glucose flavonoid 3-O-glucosyltransferase (UGFT) and glutathione S-transferase (GST) (Borevitz et al. 2000; Gonzalez et al. 2008). These enzymes are relatively conserved among plant species. These structural genes of anthocyanin biosynthesis are regulated by a complex comprising R2R3-Myeloblastosis (MYB), basic helix-loop-helix (bHLH), and tryptophan-aspartic acid repeat (WDR) proteins (MBW) (Xu et al. 2015). Notably, R2R3-MYB transcription factors (TFs) are the primary component in the MBW complex (Davies et al. 2012).

MYBs, a large transcription factor family, involved in regulating phenylpropanoid pathway structural genes have been identified. Among them, many MYB members share the conserved MYB DNA-binding domain at the N-terminus (Wu et al. 2022). The domain generally includes up to four about 52 amino acid sequence repeats (R), each forming three α -helices (Ogata et al. 1996). According to the number of adjacent MYB repeats, MYBs can be classified into several different subfamilies (Du et al. 2012). R2R3-MYBs are the most common in the plant kingdom (Jiang et al. 2004). In pear, there were 231 MYB proteins. Among them, 185 typical R2R3-MYB proteins (Li et al. 2016). Until now, several PyMYBs are involved in regulating proanthocyanin, flavonoid, and anthocyanin accumulation in pears (Zhai et al. 2016; Yao et al. 2017). For example, PyMYB114 acts as an activator to regulate anthocyanin accumulation when it couples with PybHLH3 and *Pyrus ethylene response factor 3* (PyERF3) (Yao et al. 2017). A TT2-type MYB PyMYB9 not only could bind to the PyANR promoter to induce pro-anthocyanidin synthesis, but also activate the synthesis of flavonols and anthocyanins by binding to the PyUGFT1 promoter (Zhai et al. 2016). PyMYB10 physically interacts with PyHY5 to positively regulate anthocyanin biosynthesis in pear by binding to G-box motifs (Wang et al. 2020). In other species, it was reported that the association of MdMYBA or MdMYB1 and MdMYB10 regulated anthocyanin biosynthesis in *Rosaceae* (Lin et al. 2010). AtMYB113, AtMYB75, and AtMYB114 in *Arabidopsis*, VvMYBA1 and VvMYBA2 in grapevine (*Vitis vinifera* L.), PpMYB10.1 as activators have been reported to regulate flavonoid, anthocyanin, and proanthocyanin biosynthesis in peach (*Prunus persica*) (Rahim et al. 2014).

Besides MYB activators, MYB TFs also acted as a negative regulator in anthocyanin accumulation. Such inhibitors prevent excessive or inappropriate anthocyanin accumulation (Ma and Constabel 2019). There are two different MYB types that inhibit anthocyanin biosynthesis: R2R3-MYB and R3-MYB repressors, which have one or two repeats of the MYB domain region respectively (Albert et al. 2014). The R2R3-MYB repressors are identified by two conserved domains of C1 (LlSrGIDP_x T_N HR) and C2/EAR motif (LxLxL or DLNxxP) in their C terminus essential for their repressive activity (Kranz et al. 1998; Kagale and Rozwadowski 2011). The C2/EAR motif is one of the most dominant repression motifs identified in plants. Such MYB

repressors including the FaMYB44.2 (R2R3-MYB) in strawberry (*Fragaria vesca*), CmMYB7 (R3-MYB) in chrysanthemum (*Chrysanthemum morifolium* Ramat.), StMYB44 (R2R3-MYB) in potato (*Solanum tuberosum* L.) and PhMYB4 and PhMYB17 in petunia (*Petunia xatkinsiana*) (Albert et al. 2011; Wei et al. 2018; Xiang et al. 2019; Zhou et al. 2019). These TFs could regulate anthocyanin biosynthesis by binding sites of bHLH and/or anthocyanin structural genes. However, it is not clear whether these two motifs are the sole determinants of inhibitory activity since the TLLFR motif at the C terminal of *Arabidopsis thaliana* MYBL2 (AtMYBL2) has also been reported to have inhibitory activity (Matsui et al. 2008). In contrast to MYBL2, CPC and TRY did not contain any repression motif in the C-terminal region, but negatively regulated flavonoid accumulation (Tominaga et al. 2007; Wang et al. 2008; Wester et al. 2009). In red-skinned pear, it is unclear whether or not MYBs as repressors to regulate anthocyanin accumulation.

Hydrogen sulfide (H₂S), a gaseous signaling molecule, has important effects on physiological functions, such as root formation, flower opening, stress response, and color change during fruit ripening (Yao et al. 2018; Shen et al. 2020; Ma et al. 2021; Zhang et al. 2021). In tomatoes (*Solanum lycopersicum*), H₂S could repress the fruit ripening process and color change by the H₂S production enzyme SILCD1 (Hu et al. 2020). H₂S-mediated persulfidation of the target protein is considered to be a type of post-translational protein modification. In *Arabidopsis*, it was reported that nearly 5% of proteins were modified by replacing the thiol group (-SH) at cysteine residues with the persulfidation group (-SSH), and most of the persulfidation proteins change the structure and function in activity or subcellular localization. Accumulative evidences showed that H₂S control autophagy, stomatal closure, and response stress by persulfidating modification (Laureano-Marín et al. 2020; Shen et al. 2020; Zhang et al. 2021; Zhou et al. 2021); In horticultural crops, H₂S persulfured BraFLC1/3 to inhibit its binding activity to downstream target gene promoters, resulted in the flowering process in Chinese cabbage (Ma et al. 2021). Thus, H₂S-mediated persulfidation involved in many aspects of plant development, but the post-translational modification whether or not regulated anthocyanin accumulation in red-skinned pear remains unclear.

In this study, to explore the relationship between H₂S and anthocyanin accumulation in red-skinned pear, exogenous H₂S treated 'Aohong No 1' red-skinned pears during the fruit ripening and analyzed anthocyanin contents and gene expressions that responded to H₂S signaling, two MYB genes PyMYB73 with partial EAR motif (LxLxP) and PyMYB6 with complete EAR motif (LxLxL) were screened. Co-expression of PyMYB73/PyMYB6 with PyMYB10-PybHLH3 in strawberry and pear fruits, anthocyanin contents markedly increased/decreased by transient transformation system. To elucidate the finely regulatory pattern of H₂S-mediated the post-transcriptional modifications of anthocyanin-related proteins, we identified the modification site of PyMYB10, PyMYB73, and PyMYB6 by modified biotin switching methods, western blot, and LC-MS/MS.

Interestingly, it was found that PyMYB10 exists in sulfhydrylation modification at Cys194 and Cys218. Transient expression system combined with amino acid mutation technology proved that PyMYB10^{Cys218Ala} S-sulfhydrylation substantially inhibits anthocyanin accumulation in strawberry and pear fruits, and it also promoted the interaction with PyMYB6, but weakened the interaction with PyMYB73 by Y2H and luciferase complementation experiment. Thus, our findings provided a molecular mechanism for H₂S-mediated anthocyanin biosynthesis in red-skinned pears.

Results

PyMYB73 and PyMYB6 were screened by exogenous H₂S-treated ‘Aohong No.1’ and transcriptome of ‘Hongzaosu’ and ‘Zaosu’ pear

To investigate the relationship between gaseous signal molecule H₂S and the coloration of red-skinned pear, distilled

water and NaHS solutions (as the donor of H₂S) of different concentrations (0.3, 0.6, 0.9 mM) were used to treat bagged ‘Aohong No.1’ pear without color during the fruit mature period, and three different treatment times (1, 2, 3 d) and three different light times (3, 6, 9 d) were set. The result showed that compared with the control group (CK), the fruit gradually accumulated anthocyanin with the extension of light time. At the same time, with the increase of H₂S treatment concentration, the degree of pigment deposition in the fruit epidermis decreased continuously, and the change of anthocyanin contents was largely, especially at the light irradiation for 9 d (L9D), anthocyanin contents were two times in 0.9 mM for 3 d H₂S than those of CK ($P < 0.05$) (Fig. 1, A and B). In addition, we detected the endogenous H₂S production after dark treatment with H₂S at different concentrations for 2 d and then irradiated for 9 d using the lead sulfide method. As shown in Supplemental Fig. S1A, with the increase of H₂S concentration, the color on the lead

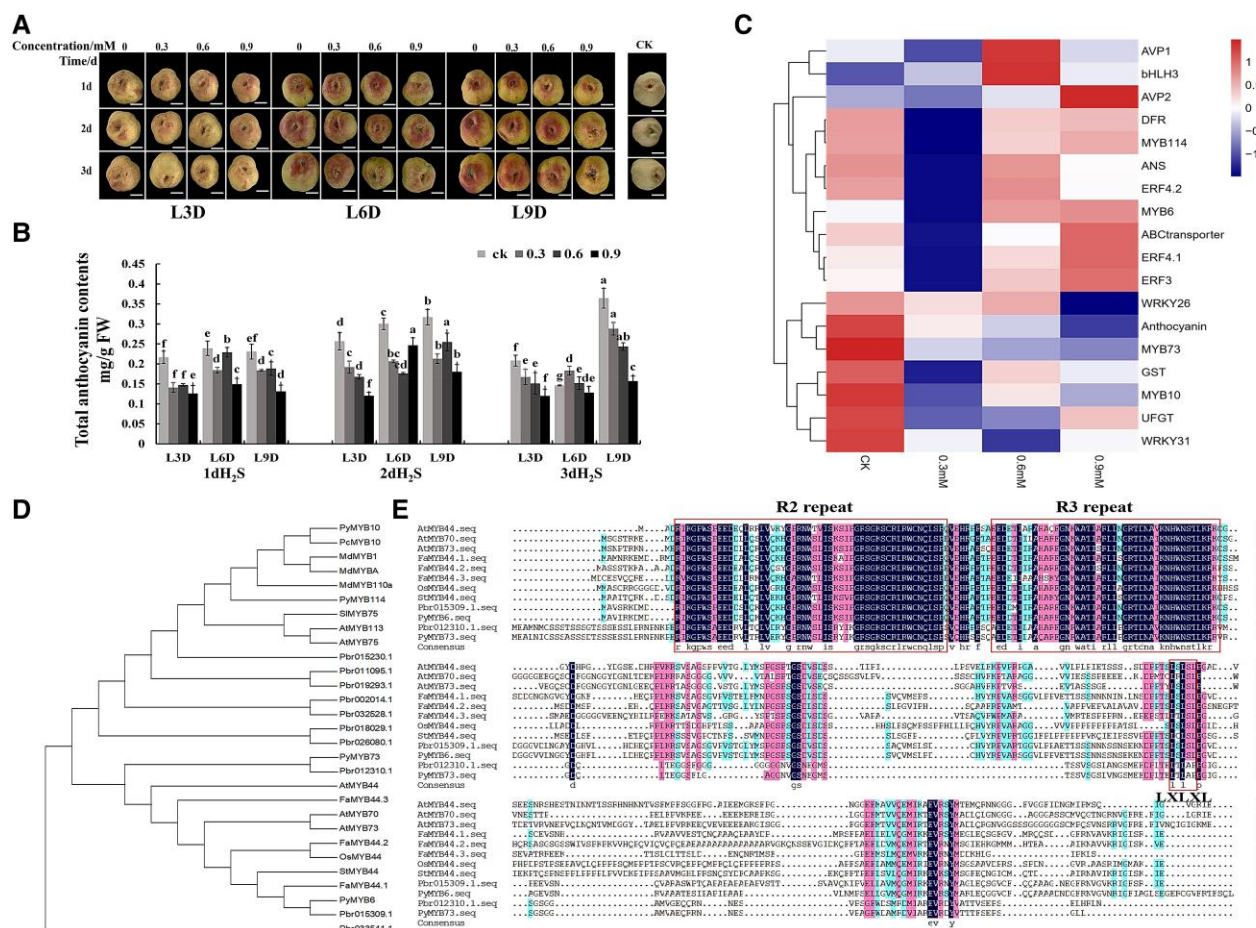


Figure 1. Screening of candidate MYB genes responding to H₂S signal in red-skinned pears. **A**) The phenotypic results of ‘Aohong No.1’ pear fruits treated with different concentrations of H₂S (0.0, 0.3, 0.6, and 0.9 mM) for 1, 2, and 3 d and then lighted for 3, 6, and 9 d. The scale bars in the figure represent 2 cm. **B**) Determination of the total anthocyanin contents. Error bars represent the mean values $\pm$ SD from three biological replicate reactions. Statistical analysis was carried out with Student’s *t*-test or Duncan’s multiple range tests, bars with different letters are significantly different at the $P < 0.05$ level. **C**) Analysis of the relationship between the expression pattern of transcription factors that are related to anthocyanin synthesis and the anthocyanin content with different concentrations of H₂S for 2 d and lighting for three days. **D**) The phylogenetic tree of MYB TFs was constructed from pears and other species. **E**) Amino acid sequence analysis of MYB TFs from different species which regulated anthocyanin biosynthesis. ‘LxLxL’ indicates the EAR motif.

acetate filter paper gradually deepened. Subsequently, the Luminance (L^*) (Supplemental Fig. S1B) and color change (a^*/b^*) (Supplemental Fig. S1C) of the lead acetate filter paper were measured via colorimeter, and similar results were obtained ($P < 0.05$). In addition, the anthocyanin contents of the sample were measured and found that its trend was contrary to the generation of endogenous H_2S ($P < 0.05$) (Supplemental Fig. S1D). Thus, the above results suggest that H_2S could inhibit anthocyanin biosynthesis in a concentration-dependent manner in red-skinned pears.

To further explore the effect of H_2S on anthocyanin synthesis in red-skinned pear, we screened a series of differentially expressed transcription factors involved in anthocyanin biosynthesis by analyzing the transcriptome data of 'Hongzaosu' and 'Zaosu' at 60 d after full blooming (DAFB), which listed in Supplemental Table S1. Then these differentially expressed genes with known MYB TFs in pears and other species were analyzed through the evolutionary tree and amino acid residue sequence alignment. *PyMYB73* and *PyMYB6* are related to *FaMYB44.2* by a phylogenetic analysis, which acts as a transcriptional repressor in strawberries (Fig. 1D). Thus, we speculated that these *PyMYB73* and *PyMYB6* may also involve in regulating anthocyanin biosynthesis in pear. Then, we analyzed the derived amino acid alignment of *PyMYB73*, *PyMYB6*, and other MYB genes related to the anthocyanin biosynthesis, such as MYB TFs, revealing the presence of R2R3 or R3 domain in all these genes, *PyMYB6* has complete EAR motif (LxLxL) whereas *PyMYB73* include an uncomplete amino acid residue (LxLxP) with the phenylalanine "P" replaced to the leucine "L" (Fig. 1E). In addition, we found that the close relatives of Pbr012310.1 and Pbr015309.1 screened from the transcriptome were *PyMYB73* and *PyMYB6*, respectively. Therefore, we performed RT-qPCR to analyze the four candidate genes and other genes related to anthocyanin synthesis in 'Zaosu' and 'Hongzaosu' at different stages after flowering and found that the expression patterns of genes related to anthocyanin synthesis in Pbr012310.1 and Pbr015309.1 were different. However, the expression pattern of *PyMYB73* is similar to that of *PyMYB10*, *PyMYB114*, and *PybHLH3*, which have been reported to regulate anthocyanins, while *PyMYB6* was opposite (Supplemental Fig. S2). It may imply that *PyMYB73* and *PyMYB6* have different functions in anthocyanin biosynthesis. To further explore the relationship between H_2S with anthocyanin biosynthesis-related genes, we also performed the RT-qPCR in the red-skinned pears from H_2S treated for 2 d under light for 3 d. The expression levels of transcription factors and structural genes such as *PyMYB114*, *PyMYB10*, *PyWRKY26*, *PyANS*, *PyDFR*, and *PyUFGT*, which have been reported to be related to anthocyanin biosynthesis, were determined and heatmaps were analyzed. It was found that the expression pattern of *PyMYB73* was similar to that of *PyMYB10* and anthocyanin contents, while that of *PyMYB6* was the opposite (Fig. 1C). It also showed that *PyMYB73* and *PyMYB6* could respond to H_2S signal and play different roles in regulating the biosynthesis of anthocyanins.

Co-transformation of *PyMYB73/PyMYB6* with *PyMYB10/PyMYB114* thereby affecting anthocyanin synthesis in strawberry receptacles

To prove the function of *PyMYB73* and *PyMYB6* in anthocyanin synthesis, we carried out a transient expression assay in strawberry receptacles via agroinfiltration. As shown in Fig. 2A, the pSAK277 was injected as the negative control, when *PyMYB73/PyMYB6* alone was injected, which did not show any pigmentation. Co-injection of *PyMYB10* or *PyMYB114* with *PybHLH3* could be observed weak pigmentation, and obvious color change could be observed when *PyMYB10-PybHLH3-PyMYB73* were co-injected. Similar pigmentation could be observed when *PyMYB114-PybHLH3-PyMYB73* were co-transformed. However, the accumulation of anthocyanin was substantially lower when *PyMYB10-PybHLH3-PyMYB6* or *PyMYB114-PybHLH3-PyMYB6* were co-injected. The total anthocyanin content of co-injecting *PyMYB10/PyMYB114-PybHLH3-PyMYB73* in a strawberry was significantly higher than that with only the co-transformation of *PyMYB10/PyMYB114* and *PybHLH3* ($P < 0.05$). However, when *PyMYB6* was co-injected with *PyMYB10/PyMYB114-PybHLH3*, the opposite results for the anthocyanin contents were obtained ($P < 0.05$) (Fig. 2B). Besides, the changes of a^* value were affected by the pigmentation of the strawberry receptacles ($P < 0.05$) (Fig. 2C). Overall, co-transformation of *PyMYB73* with *PyMYB10/PyMYB114-PybHLH3* can obviously enhance the anthocyanin biosynthesis, whereas *PyMYB6* weakens that in strawberry receptacles.

To appraise the role of *PyMYB73* and *PyMYB6* in regulating the anthocyanin biosynthesis, we analyze four structural genes *FvDFR*, *FvANS*, *FvUFGT*, and *FvGST*, which are related to anthocyanin biosynthesis in strawberry receptacles by RT-qPCR. As shown in Fig. 2, D to G, cotransformation of *PyMYB10/PyMYB114-PybHLH3-PyMYB73* significantly promoted the expression levels of *FvDFR*, *FvANS*, *FvUFGT* and *FvGST* ($P < 0.05$). However, *PyMYB6* had a strong inhibitory role. We concluded that co-transformation of *PyMYB73-PyMYB10/PyMYB114-PybHLH3* can promote the anthocyanin synthesis by upregulating the expression of the *FvDFR*, *FvANS*, *FvUFGT*, and *FvGST*, but the effect of *PyMYB6* is just the opposite, which is consistent with the previous results.

PyMYB73/PyMYB6 regulates anthocyanin accumulation by competitive combination the regulatory complex of *PyMYB10/PyMYB114-PybHLH3*

To further identify the role of *PyMYB73/PyMYB6* in anthocyanin biosynthesis, different ratios of *PyMYB73/PyMYB6* (1:1/2:1/1:2) with *PyMYB10/PyMYB114* and *PybHLH3* were co-transformed into pear peels. We performed the transient transformation in mature pear fruits at 90 DAFB. The pigmentation in fruit peels could be observed at the injection regions at 5 d postinfiltration. The pattern of change was similar to that of strawberry receptacles. As shown in Fig. 3A, no pigmentation was observed when the pSAK277 was infiltrated. Apparent pigmentation accumulation could

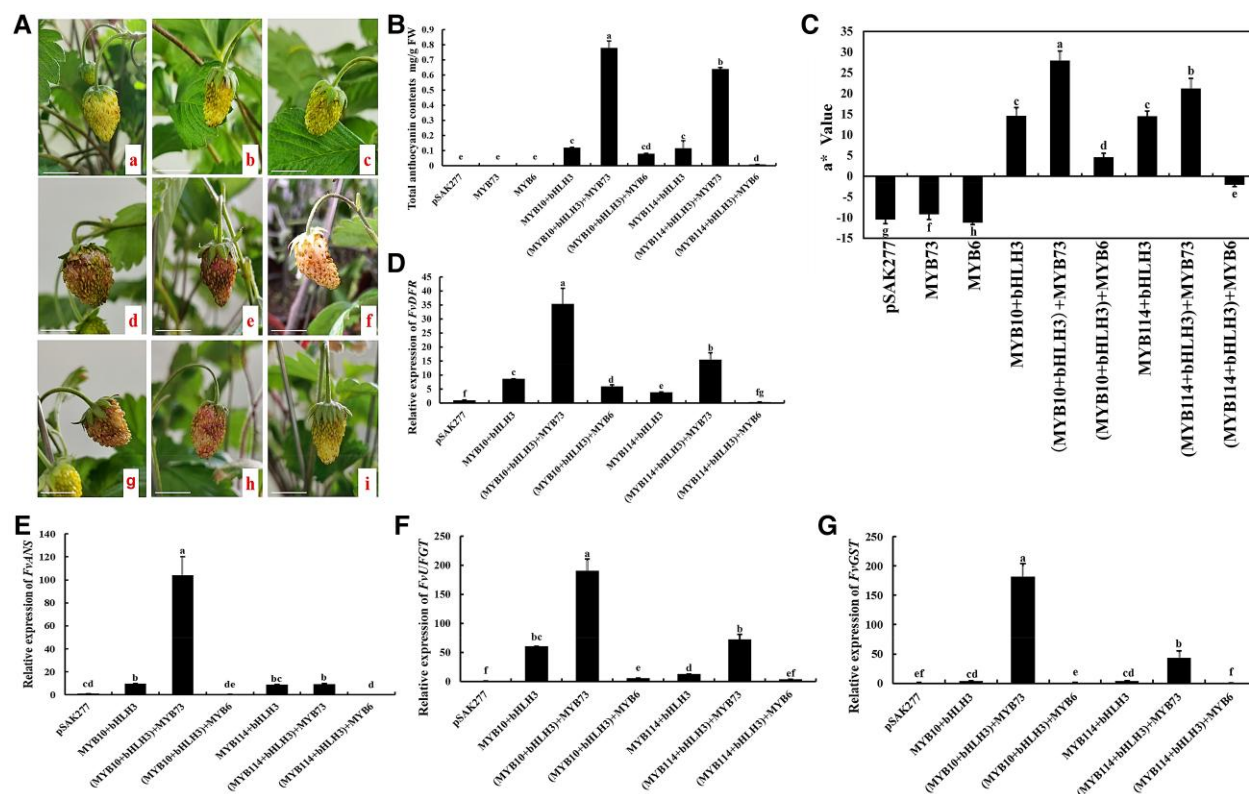


Figure 2. Coinjection of PyMYB73/PyMYB6 and related TFs by transient expression assay in strawberry receptacles. **A)** The phenotype of strawberry receptacles 5 d after infiltration: a, pSAK277; b, PyMYB73; c, PyMYB6; d, PyMYB10 + PybHLH3; e, PyMYB10 + PybHLH3 + PyMYB73; f, PyMYB10 + PybHLH3 + PyMYB6; g, PyMYB114 + PybHLH3; h, PyMYB114 + PybHLH3 + PyMYB73; i, PyMYB114 + PybHLH3 + PyMYB6. The scale bars represent 1 cm. **B)** Total anthocyanin contents were measured in transformed strawberry receptacles. Data are presented as means $\pm$ SD ($n = 5$). **C)** Measurement of the chromatism changes of the injecting regions in strawberry receptacles. "a" represents the range from red to green. Data are presented as means $\pm$ SD ($n = 6$). **D–G)** RT-qPCR analysis of the relative expression of FvDFR, FvANS, FvUGT, and FvGST. Error bars show the mean values $\pm$ SD ($n = 3$). Statistical analysis was carried out with Student's *t*-test, and significance was marked with different letters ($P < 0.05$).

be observed when PyMYB73-PyMYB10/PyMYB114-PybHLH3 were co-injected. Furthermore, the results showed that an obvious color change could be observed when the PyMYB73/PyMYB6 ratio increased from 1:1 to 2:1. However, PyMYB6 could weaken pigmentation accumulation when it was co-transformed with PyMYB10/PyMYB114 + PybHLH3 in pear fruit, and the pigmentation accumulation diminished with the decreasing ratio of PyMYB73/PyMYB6 (1:1 to 1:2). As the proportion of PyMYB6 continues to increase, the total anthocyanin contents and a^* values declined significantly ($P < 0.05$). This is the opposite of the role of PyMYB73, and their changes were in accordance to the phenotypes of the pear peel mentioned above (Fig. 3, B and C). Thus, when co-transformed with PyMYB73/PyMYB6 and PyMYB10/PyMYB114 + PybHLH3, a transcriptional complex may be formed to competitively regulate anthocyanin accumulation.

To further explore the mechanism of PyMYB73 and PyMYB6 in regulating anthocyanin synthesis. RT-qPCR was conducted to analyze the expression of PyDFR, PyANS, PyUGT, and PyGST, which were related to anthocyanin biosynthesis and transport in pear fruit. Consistent with the phenotypes of the pear peel, PyMYB73 could significantly

increase the relative expression of PyDFR, PyANS, PyUGT, and PyGST when it was co-transformed with PyMYB10/PyMYB114-PybHLH3, and as the proportion of PyMYB73 increased, this effect was continuously enhanced ($P < 0.05$). However, PyMYB6 could inhibit the relative expression levels of PyDFR, PyANS, PyUGT, and PyGST when it was mixed with PyMYB73 and PyMYB10/PyMYB114-PybHLH3, and the inhibitory effect showed a certain concentration dependence ($P < 0.05$) (Fig. 3, D to G). Furthermore, we designed transient expression experiments to verify whether the interaction between PyMYB73 and PyMYB6 affected the biosynthesis of anthocyanins in pears. As shown in Supplemental Fig. S3A, compared with pSAK277, the transformation of PyMYB73 could substantially promote the accumulation of anthocyanin, while PyMYB6 had no obvious staining. In addition, anthocyanin accumulation existed when PyMYB73 and PyMYB6 were co-transformed at different ratios (1: 1/2:1/1:2). When PyMYB73 and PyMYB6 were injected at the ratio of (2:1), the anthocyanin accumulation was significantly higher than that of (1:1/1:2). When PyMYB73 and PyMYB6 were injected at the ratio of (1:2), the accumulation of anthocyanin was significantly weaker than that of (1:1/2:1). As the

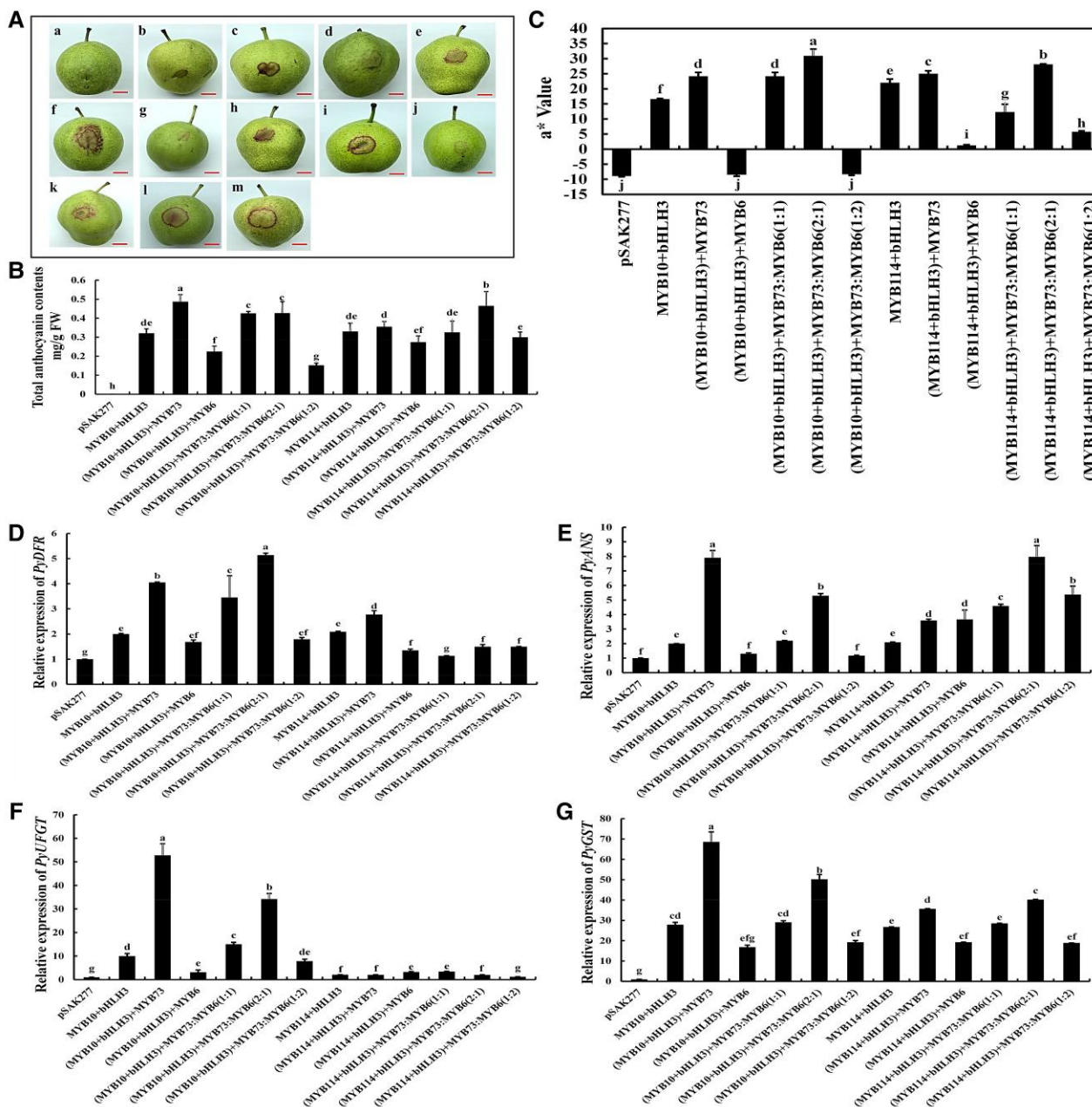


Figure 3. The functional analysis of PyMYB73/PyMYB6 cotransformed with its partners thereby effecting anthocyanin biosynthesis by transient expression assays in pear skins. (a) the phenotype changes of ‘Zaosu’ pears at 7 d after infiltration; a, pSAK277; b, PyMYB10 + PybHLH3; c, (PyMYB10 + PybHLH3) + PyMYB73; d, (PyMYB10 + PybHLH3) + PyMYB6; e, (PyMYB10 + PybHLH3) + PyMYB73:PyMYB6(1:1); f, (PyMYB10 + PybHLH3) + PyMYB73:PyMYB6(2:1); g, (PyMYB10 + PybHLH3) + PyMYB73:PyMYB6(1:2); h, PyMYB114 + PybHLH3; i, (PyMYB114 + PybHLH3) + PyMYB73; j, (PyMYB114 + PybHLH3) + PyMYB6; k, (PyMYB114 + PybHLH3) + PyMYB73:PyMYB6(1:1); l, (PyMYB114 + PybHLH3) + PyMYB73:PyMYB6(2:1); m, (PyMYB114 + PybHLH3) + PyMYB73:PyMYB6(1:2). The scale bars in the figure represent 2 cm. **B)** The total anthocyanin contents were determined in pear peels. Error bars represent the mean values $\pm$ SD from five replicate reactions. **C)** Determination of color change in pear peels by chromatism analysis. “a” represents the range from red to green. Error bars represent the mean values $\pm$ SD from six replicate reactions. **D–G)** The relative expression levels of PyDFR, PyANS, PyUFGT, and PyGST were analyzed by RT-qPCR. Error bars show the mean values $\pm$ SD ($n = 3$). Statistical analysis was carried out with Student’s *t*-test, and significance was marked with different letters ($P < 0.05$).

proportion of PyMYB73 continues to increase, the total anthocyanin contents and a^* values increased substantially ($P < 0.05$), and there was an opposite result when the proportion of PyMYB6 continues to increase ($P < 0.05$) (Supplemental Fig. S3, B and C). In addition, we

analyzed the expression of PyDFR, PyANS, PyUFGT, and PyGST by RT-qPCR. Consistent with the phenotypes of the pear peel, PyMYB73 could significantly increase the relative expression levels of PyDFR, PyANS, PyUFGT, and PyGST. When it was co-transformed with PyMYB6, and with the

proportion of PyMYB73 increased, this effect was continuously enhanced ($P < 0.05$). However, PyMYB6 could inhibit the relative expression levels of PyDFR, PyANS, PyUGT, and PyGST. When it was co-transformed with PyMYB73 the inhibitory effect showed a certain concentration dependence ($P < 0.05$) (Supplemental Fig. S3, D to G).

Identify the interaction between PyMYB73, PyMYB6 and other genes by Y2H

Cloning PyMYB73/PyMYB6 as a coding sequence (CDS), or with amino acid residues removal of the C- or N-terminal, and inserting into pGBKT7. The CDS of PyMYB10, PyMYB73, PyMYB114, and PybHLH3 were also recombined into pGADT7 to test its interaction with PyMYB73/PyMYB6. The results indicated that the full sequences (III) and the C-terminal 69 (residues 170 to 239; IV) of PyMYB73 represented reporter activity, with two N-terminal fragments, PyMYB73^{1–100} (I) and PyMYB73^{1–170} (II), and the C-end fragment PyMYB73^{100–239} (V), representing no reporter activity. When the co-conversion of the C-terminal 139 amino acids (V) of PyMYB73 with PyMYB10 representing that both growth on double deficiency and four deficiency screening media. Therefore, the

interaction site of PyMYB73 and PyMYB10 was within the N-end 139 amino acids (V) (Fig. 4, A and B). When PyMYB73 with PyMYB114 were co-converted into yeast, except for the complete amino acid sequences (III) and the C-end 69 amino acids (residues 170 to 239; IV), no other reported activity was represented (Supplemental Fig. S4A). The same result occurs when PyMYB73 and PybHLH3 were co-converted into yeast (Supplemental Fig. S4B). Moreover, seven fragments of PyMYB6 (I–VII) showing no reporter activity. The co-conversion of the N-terminal 117 amino acids (II) of PyMYB6 with PyMYB10 showed both growth on double deficiency and four deficiency screening media. Thus, the interaction sites of PyMYB6 and PyMYB10 in the regions of the N-terminal 117 amino acids (II) (Fig. 4, C and E). In addition, when the complete amino acid sequences of PyMYB6 with PyMYB73 were co-transformed into yeast, it showed reporter activity on double deficiency media and four deficiency screening media (Fig. 4D). Co-conversion of the amino acid sequences (I)–(VII) of PyMYB6 with PyMYB114 and PybHLH3 did not obvious reporter activity on four deficiency screening media. It suggested that PyMYB6 was unable to interact with PyMYB114 or PybHLH3 at the protein level (Supplemental Fig. S4, C and D). In summary, it was

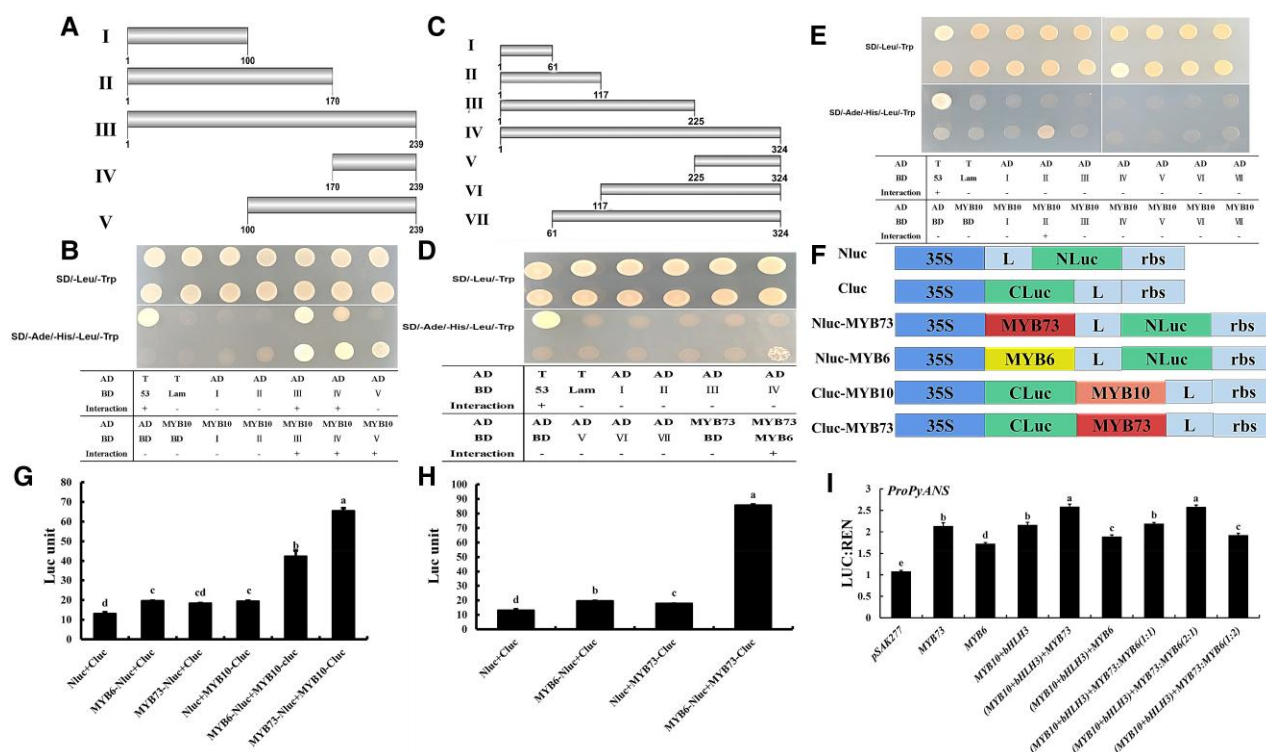


Figure 4. Verifying the regulatory pattern of PyMYB73/PyMYB6 and PyMYB10 *in vivo*. **A)** The roman numerals I–V indicated different amino acid residues of PyMYB73. **B)** Validation of interaction between PyMYB73 and PyMYB10 *in vivo*. **C)** The roman numerals I–VII indicated different amino acid residues of PyMYB6. **D)** Validation of interaction between PyMYB73 and PyMYB6 *in vivo*. **E)** Validation of interaction between PyMYB6 and PyMYB10 *in vivo*. **F)** Model of the NLuc, CLuc, and NLuc/CLuc constructs. **G and H)** The interactions between PyMYB10, PyMYB6, and PyMYB73 were verified using leaflet *Nicotiana benthamiana*. Bars represent mean values $\pm$ SD from five biological replicates. **I)** Verifying of PyMYB10, PybHLH3, and PyMYB73 or PyMYB6 cotransformation activates the PyANS promoters, and the LUC/REN value indicated promoter activity by the dual luciferase reporter (DLR) assay. Error bars show the mean values $\pm$ SD ($n = 5$). Statistical analysis was carried out with Student's *t*-test, and significance was marked with different letters ($P < 0.05$).

concluded that PyMYB73/PyMYB6 can interact with PyMYB10, and PyMYB73 also can interact with PyMYB6 in yeast.

To further prove the interaction between PyMYB10 and PyMYB73/PyMYB6, the luciferase complementary experiment was conducted. PyMYB73/PyMYB6 was recombined into the firefly luciferase (NLuc)'s N-terminal regions, and PyMYB10 and PyMYB73 were inserted into the firefly luciferase (CLuc)'s C-terminal regions (Fig. 4F). Co-expression of the NLuc-PyMYB73 or NLuc-PyMYB6 and CLuc-PyMYB10 recombination indicated significant luciferase enzyme activity ($P < 0.05$). Co-expression of the NLuc-PyMYB6 and CLuc-PyMYB73 recombination indicated obvious luciferase enzyme activity as well ($P < 0.05$). On the contrary, there was no apparent luciferase enzyme activity in any control groups (Fig. 4, G and H). In general, it was concluded that there existed an interaction between PyMYB73, PyMYB6, and PyMYB10. These implied that a transcriptional regulatory complex of PyMYB73-PyMYB6-PyMYB10 regulated anthocyanin biosynthesis in pear.

The dual luciferase reporter assay (DLR) the cotransformation of PyMYB73/PyMYB6 affecting the activity of PyDFR, PyANS, and PyUGT promoters

Performing a DLR in *Nicotiana benthamiana* to confirm the activation for PyDFR, PyANS, and PyUGT via co-injecting PyMYB73/PyMYB6-PyMYB10-PybHLH3. The results indicated that PyMYB73/PyMYB6 effected these promoters' activation. Co-transformation of PyMYB73 or PyMYB6 could improve the activation of the PyANS promoters with PyMYB10-PybHLH3 ($P < 0.05$), and there is a significant difference in co-transformation of various ratios of PyMYB73/PyMYB6 (1:1/2:1/1:2) and PyMYB10 +PybHLH3 ($P < 0.05$) (Fig. 4I). However, PyMYB73 with PyMYB6 co-transformation showed no significant activation to PyDFR and PyUGT (Supplemental Fig. S4, E and F). It appears that PyMYB73 and PyMYB6 can form regulatory complex PyMYB73-PyMYB10-PybHLH3-PyMYB6 to regulate the transcription of anthocyanin biosynthetic genes PyDFR and PyUGT.

Identification of persulfidation modification at Cys194 and Cys218 of PyMYB10

From the above analysis, we found that PyMYB73 and PyMYB10 showed similar expression patterns after being treated with different concentrations of exogenous H_2S for 2 d and exposed to light for 2 d, both of which were down-regulated with the increase of treatment concentration, but the expression level of PyMYB6 showed was opposite. At the same time, we found that PyMYB73 and PyMYB6 played a different role in regulating anthocyanin synthesis in strawberry and pear fruits. We speculate that the reason for this phenomenon may be the proteins were modified persulfidation through H_2S . First, we expressed and purified PyMYB73, PyMYB10, and PyMYB6 proteins from *Escherichia coli*. The expression and purification results of

PyMYB10 are shown in Fig. 5, A and B. The expression and purification results of PyMYB6 and PyMYB73 are shown in Supplemental Fig. S5, A to D. Then we used a modified biotin-switch method (MBSM) to detect the signal of persulfidation (Chen et al. 2021). As shown in Fig. 5, C and D, NaHS induced persulfidation of PyMYB10 in a dose-dependent manner, and when dithiothreitol (DTT) was added, which could eliminate the persulfidation, we did not observe any signals of persulfidation in PyMYB73 and PyMYB6 (Supplemental Fig. S5, E and F). To further verify this result, we performed liquid chromatography–tandem mass spectrometry (LC–MS/MS) to analyze the recombinant PyMYB10 protein. It was determined that the Cys194 and Cys218 residues had acquired a sulfhydryl modification (Fig. 5, E and F).

Mutations at Cys194 and Cys218 of PyMYB10 can positively regulate the biosynthesis of anthocyanins in strawberry receptacles

To determine whether the two cysteine residues of PyMYB10 could affect anthocyanin synthesis, we mutated Cys194 and Cys218 of PyMYB10 to alanine (Fig. 5G). Then we carried out transient transformation in strawberry receptacles via agroinfiltration. As shown in Fig. 6A, except pSAK277, other combinations had obvious pigment deposition on the strawberry fruit epidermis, and it was worth noting that the pigment accumulation was deepened after the mutation of two cysteine residues of PyMYB10, especially PyMYB10^{Cys218Ala}. The total anthocyanin content and a* value were analyzed, which were the same as the visual observations ($P < 0.05$) (Fig. 6, B and C). To explore the mechanism of PyMYB10 in regulating anthocyanin synthesis, we detected the expression levels of anthocyanin biosynthesis-related genes by RT-qPCR analysis. It could be seen from Fig. 6, D to G, which except pSAK277, other combinations could improve the expression levels of anthocyanin biosynthesis-related genes, especially PyMYB10^{Cys194Ala}, PyMYB10^{Cys218Ala} and PyMYB10^{Cys194AlaCys218Ala} could significantly improve ($P < 0.05$). These results suggested that mutations of PyMYB10^{Cys194Ala} or PyMYB10^{Cys218Ala} would increase anthocyanins accumulation, and it seems that Cys218 might play a more important role by comparing with Cys194.

H_2S inhibits the function of PyMYB10 through persulfidation at Cys194 and Cys218 in pears

Through heterologous expression in strawberries, we preliminarily infer that the mutation of PyMYB10^{Cys194Ala} or PyMYB10^{Cys218Ala} would increase anthocyanins accumulation. To verify this result again, we performed the transient transformation in mature pear fruits (90 DAFB) and treated a group with 0.9 mM NaHS. Fruit color could be noticed at the infiltration sites after 5 d. As shown in Fig. 7A, after 0.9 mM NaHS treatment, the pigment deposition of PyMYB10 was weakened, the pigment deposition of PyMYB10^{Cys194Ala} and double

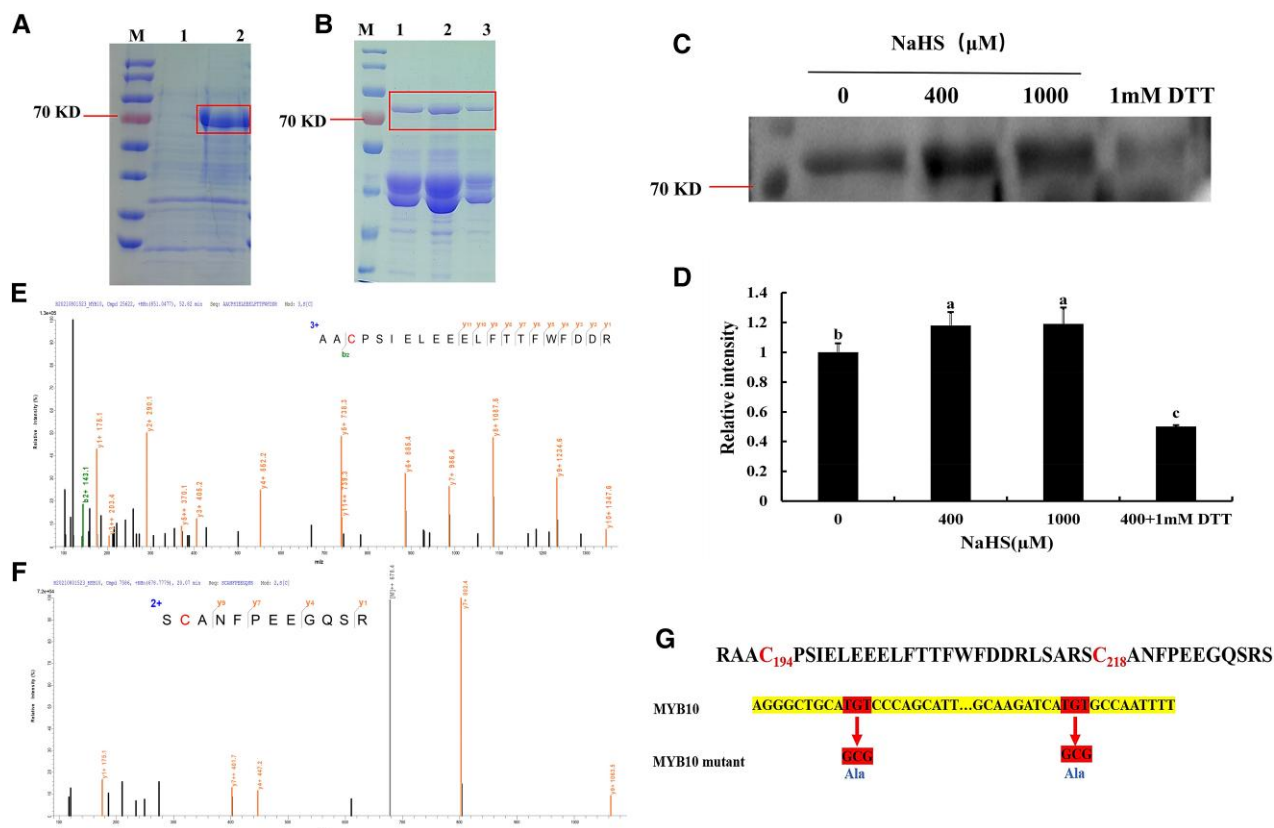


Figure 5. Identification of PyMYB10 persulfidation modification sites. **A)** The expression of PyMYB10 protein. M, marker of protein molecular weight; 1, before induction; 2, after induction. **B)** Purification of PyMYB10 protein. 1–3 indicated the number of protein elution. **C)** *In vitro* PyMYB10 persulfidation was detected using the biotin-switch assay. Proteins were treated with different concentration of NaHS or NaHS + DTT (1 mM) for 30 min. **D)** Quantification of the degree of persulfidation shown in **C)**. Error bars are presented as means $\pm$ SD ($n = 3$). Statistical analysis was carried out with Student's *t*-test, and significance was marked with different letters ($P < 0.05$). **E and F)** Mass spectrometric analysis of the persulfidation modification sites of PyMYB10 protein. LC–MS/MS spectrum implied that the peptide, RAA₁₉₄PSIELEELFTTFWFDDRL and RS₂₁₈CANFPPEGQSRS are persulfidated under control conditions. **G)** Schematic diagram of PyMYB10 sulfhydrylation modification sites by Cys–Ala. The red letters indicate sulfhydrylation of amino acids at 218 and 194 sites, respectively. The letters on the yellow background represent the CDS sequence corresponding to the amino acid sequence, where in the “TGT” on the red background represents the corresponding codon of the 194 and 218 amino acids, respectively, and the “GCC” on the red background represents the codon after the “TGT” mutation to Alanine (Ala).

mutations PyMYB10^{Cys194AlaCys218Ala} had no substantial difference, while PyMYB10^{Cys218Ala} was enhanced. The results of total anthocyanin content and a^* value supported the visual observations ($P < 0.05$) (Fig. 7, B and C). Then, we also conducted RT-qPCR to analyze the relative expression levels of anthocyanin biosynthesis-related genes in pears. The results demonstrated that in the H₂S treatment group, PyMYB10 decreased the expression levels of anthocyanin biosynthesis-related genes (PyDFR, PyANS, PyUGT, and PyGST), while after the mutation of PyMYB10^{Cys194Ala}, PyMYB10^{Cys218Ala}, and PyMYB10^{Cys194AlaCys218Ala} were injected, the expression levels of these genes increased except PyANS, especially the expression levels of PyDFR, PyUGT, and PyGST in PyMYB10^{Cys218Ala} were significantly higher than those of other combinations ($P < 0.05$) (Fig. 7, D to G). Then we conducted a DLR assay to explore the activation activity of PyMYB10 on the promoters of PyDFR, PyANS, and PyUGT. The results displayed that PyMYB10 might regulate anthocyanin synthesis by activating PyDFR ($P < 0.05$) (Supplemental Fig. S6, A to C). In addition, it was determined

the effect of H₂S in regulating PyMYB10 activating PyDFR promoter. We found that H₂S could decrease the activation activity of PyMYB10 on the promoter, PyMYB10^{Cys194AlaCys218Ala} no activation activity change, while the activation activity of PyMYB10^{Cys194Ala} and PyMYB10^{Cys218Ala} increased, especially PyMYB10^{Cys218Ala} ($P < 0.05$) (Supplemental Fig. S6D). In a word, H₂S inhibits anthocyanin synthesis by persulfidation at Cys194 and Cys218 of PyMYB10 to weaken the activation for downstream target genes.

Mutations at Cys194/Cys218 of PyMYB10 increased anthocyanin biosynthesis by co-transforming with PyMYB73/PyMYB6 in pear

Through transient expression in strawberries and pears, we found that mutations of Cys194Ala, Cys218Ala, and Cys194AlaCys218Ala in PyMYB10 could promote anthocyanin synthesis. However, for further exploring the effect of PyMYB10 mutation on its interaction factor PyMYB73/

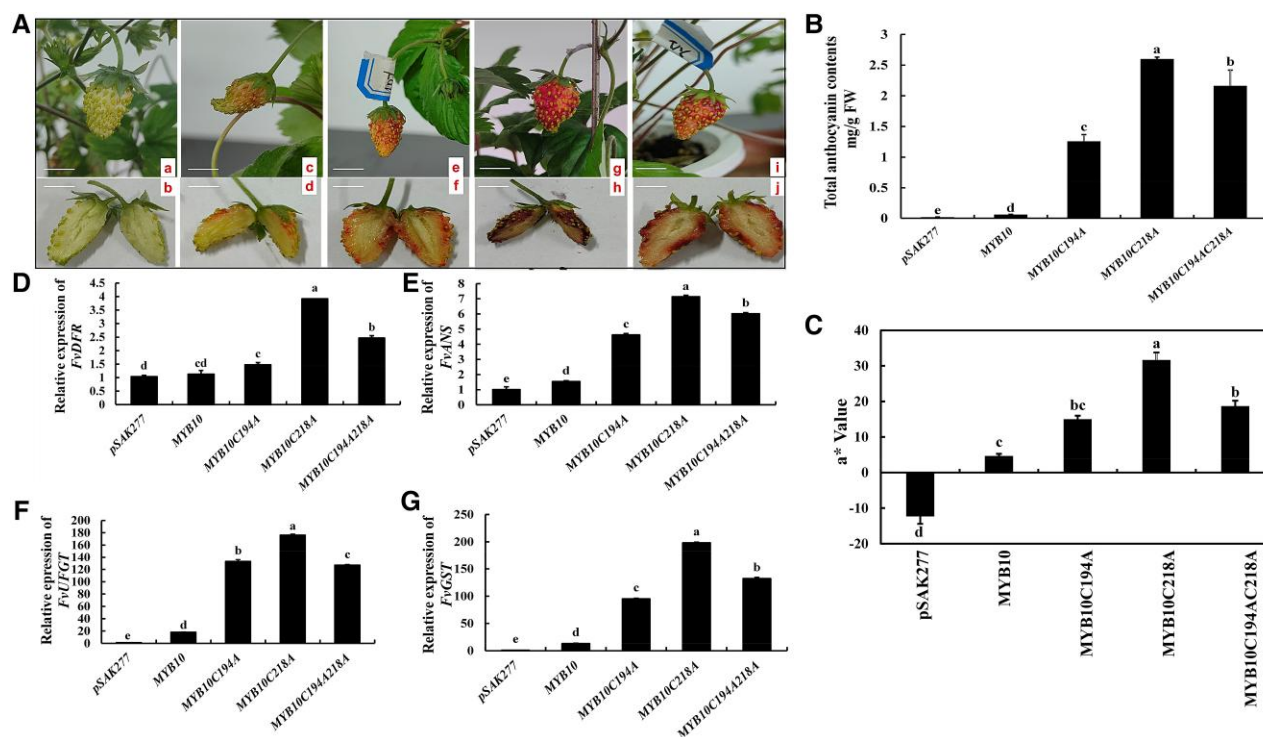


Figure 6. Effect of the two cysteine mutations of PyMYB10 on anthocyanin synthesis, by transient expression assay in strawberry receptacles. **A)** The phenotype of strawberry receptacles at 5 d after infiltration: a and b, pSAK277; c and d, PyMYB10; e and f, PyMYB10^{Cys194Ala}; g and h, PyMYB10^{Cys218Ala}; i and j, PyMYB10^{Cys194AlaCys218Ala}. The scale bars represent 1 cm. **B)** Total anthocyanin contents were determined in strawberry receptacles. Data are presented as means $\pm$ SD ($n = 5$). **C)** Chromatism analysis of the injection regions in strawberry receptacles. “a” represented the range from red to green. Data are presented as means $\pm$ SD ($n = 6$). **D–G)** RT-qPCR analysis of the expression levels of the genes, *FvDFR*, *FvANS*, *FvUGT*, and *FvGST*. Error bars show the mean values $\pm$ SD ($n = 3$). Statistical analysis was carried out with Student’s *t*-test, and significance was marked with different letters ($P < 0.05$).

PyMYB6, we co-transformed PyMYB10 mutations with PyMYB73/PyMYB6 into pear fruit for transient expression. It can be observed that all combinations showed pigment accumulation, which was higher than that of the control group. From the phenotype, total anthocyanin content, and a^* value, it can be concluded that PyMYB10 with the Cys218Ala mutation, is higher than other groups ($P < 0.05$) (Fig. 8, A to C). RT-qPCR was constructed to determine the expression patterns of anthocyanin biosynthesis-related genes. When PyMYB10^{Cys218Ala} co-transformed with PyMYB73, the expression levels of *PyDFR*, *PyANS*, *PyUGT*, and *PyGST* were higher than those of unmutated ones, as was PyMYB6 ($P < 0.05$) (Fig. 8, D to G). It suggested that the Cys218 site of PyMYB10 played an important role in affecting anthocyanin synthesis.

Validation of the interaction between PyMYB73/PyMYB6 and mutations of Cys194Ala, Cys218Ala or Cys194AlaCys218Ala in PyMYB10

The yeast two-hybrid experiments were used to verify the interaction between PyMYB10 contained the sites of Cys194Ala or Cys218Ala or Cys194AlaCys218Ala with PyMYB73/PyMYB6 segments and full-length. It can be seen from Fig. 9, A to C, after the PyMYB10^{Cys194Ala},

PyMYB10^{Cys218Ala}, and PyMYB10^{Cys194AlaCys218Ala} mutation, it still interacted with PyMYB73 that with the N-terminal 139 amino acids (V), but there is no interaction effect with PyMYB6. For further identification of the interaction between PyMYB10^{Cys194Ala}, PyMYB10^{Cys218Ala}, and PyMYB10^{Cys194AlaCys218Ala} with PyMYB73/PyMYB6 by an FLC assay in *N. benthamiana*. It can be concluded that PyMYB10 with the Cys194, Cys218, and Cys194Cys218 mutation, were inserted into the firefly luciferase (CLuc)’s C-terminal region, which co-expression with the NLuc-PyMYB73 displayed the capability to luciferase enzyme activity. However, NLuc-PyMYB6 turned out to be the opposite. In addition, we found that compared with CK, when treated with exogenous H₂S, the luciferase enzyme activity of Cluc-PyMYB10 and NLuc-PyMYB73 decreased to a certain extent, while the luciferase enzyme activity of NLuc-PyMYB6 and Cluc-PyMYB10 increased significantly ($P < 0.05$), but the luciferase enzyme activity of Cluc-PyMYB10, which contained two cysteine mutation, with NLuc-PyMYB73/PyMYB6 did not change substantially (Fig. 9C). It suggested that H₂S may enhance the interaction between PyMYB10 and PyMYB6, weaken the interaction between PyMYB10 and PyMYB73 and H₂S may inhibit anthocyanin biosynthesis by affecting the interaction between PyMYB10 and PyMYB73/PyMYB6.

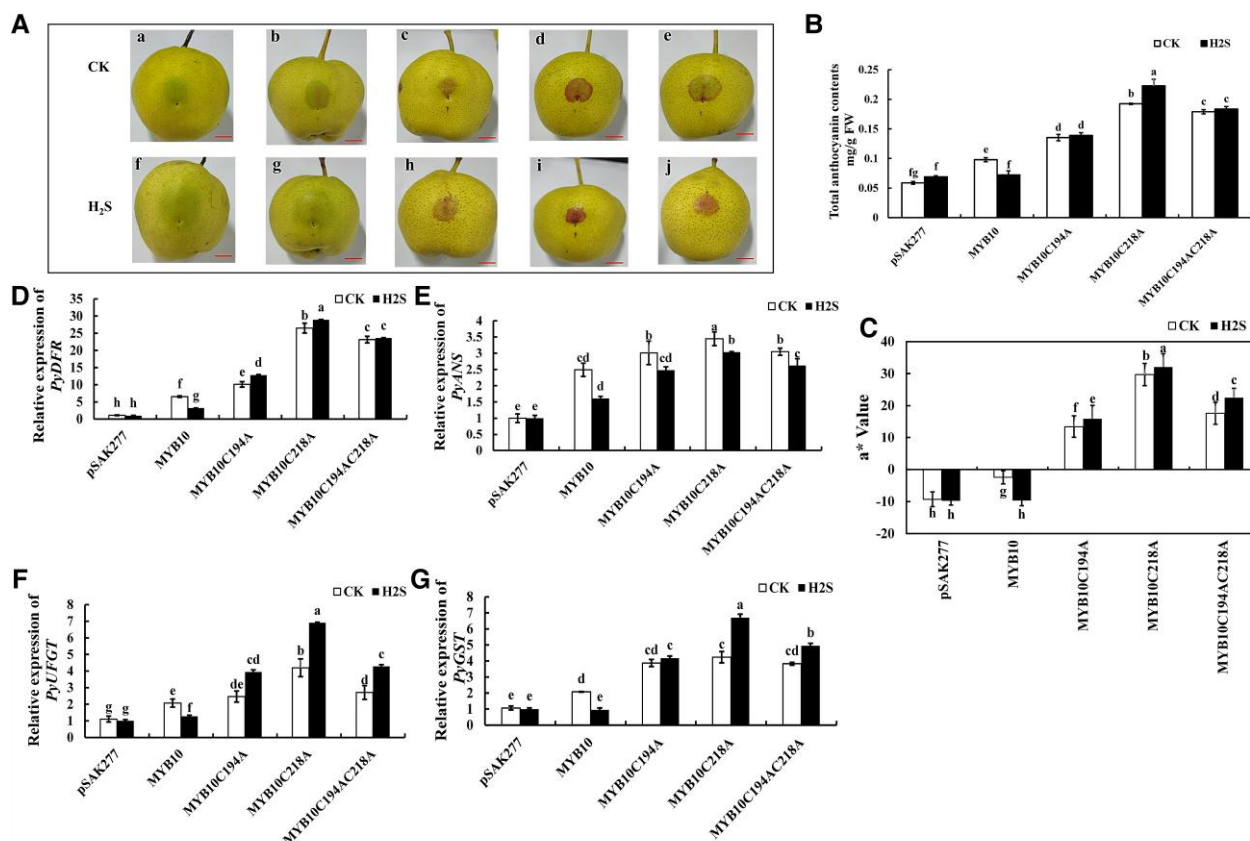


Figure 7. Effect of the two cysteine mutations of PyMYB10 on anthocyanin synthesis with H₂S treatment by transient expression assays in pears. **A)** The phenotypic results of the pear cultivar 'Zaosu' at 4 d after infiltration. CK: dark treat for 24 h and light for 4 d; H₂S: (dark and 0.9 mM NaHS) treat for 24 h and light for 4 d. a and f: pSAK277; b and g: PyMYB10; c and h: PyMYB10^{Cys194Ala}; d and i: PyMYB10^{Cys218Ala}; e and j: PyMYB10^{Cys194AlaCys218Ala}. The scale bars in the figure represent 2 cm. **B)** The total anthocyanin contents were determined in pear peels. Error bars represent the mean values ± SD from three biological replicates. **C)** Measurement of the color change in pear peels by chromatism analysis. "a" indicated the range from red to green; Error bars indicate the mean values ± SD from six biological replicates. **D–G)** RT-qPCR analysis of the expression levels of the genes, *PyDFR*, *PyANS*, *PyUGFT*, and *PyGST* in pear peels. Error bars show the mean values ± SD (n = 3). Statistical analysis was carried out with Duncan's multiple range tests, bars with different letters are significantly different at the P < 0.05 level.

Discussions

A complete EAR motif is essential to repress anthocyanin biosynthesis in red-skinned pear

Red-skinned pear has been widely cultivated and accepted by consumers around the world because of its attractive color and underlying nutritional value. However, the coloration of fruits is influenced by a series of factors, including transcription factors MYBs, ERFs, ethylene, H₂S, and other hormones (Yao et al. 2017, 2018; Tao et al. 2018; Hu et al. 2020; Ni et al. 2021). It may imply that they have different regulatory mechanisms and complex regulatory networks. In our present study, we have observed that there was dominant anthocyanin deposition in red-skinned pear 'Hongzaosu' compared with 'Zaosu' during the fruit development stages. Subsequently, a pear MYB gene, namely PyMYB73, was screened from the transcriptome data for the upregulated expression in red-skinned pear peels, which may involve in anthocyanin biosynthesis (Supplemental

Table S1). Then we found another MYB that belongs to the same branch as PyMYB73, namely PyMYB6, which contains a complete C2/EAR motif (LxLxL) located in the C terminus, and PyMYB73 has an incomplete EAR motif (LxLxP) (Fig. 1, D and E). Further identification showed that PyMYB73 could enhance anthocyanin accumulation in strawberry receptacles and pear fruit, especially when co-transformed with other TFs (Fig. 3A and 4A). However, PyMYB6 including an "LxLxL" motif resulted in weakening anthocyanin accumulation, it is different from the function of PyMYB73. The repressors usually contain a C2/EAR (LxLxL or DLNxxP) negative repressor motif located in the C-terminal region, which is essential to the repressive effect (Zheng et al. 2019). According to recent reports, flesh-specific StMYB44 represses the biosynthesis of anthocyanin in potato flesh (Liu et al. 2019). In peach, PpMYB18 as a negative R2R3-MYB regulator competitively inhibits the interaction of MYB activators with bHLHs, resulting in a decrease in anthocyanin content (Zhou et al. 2019).

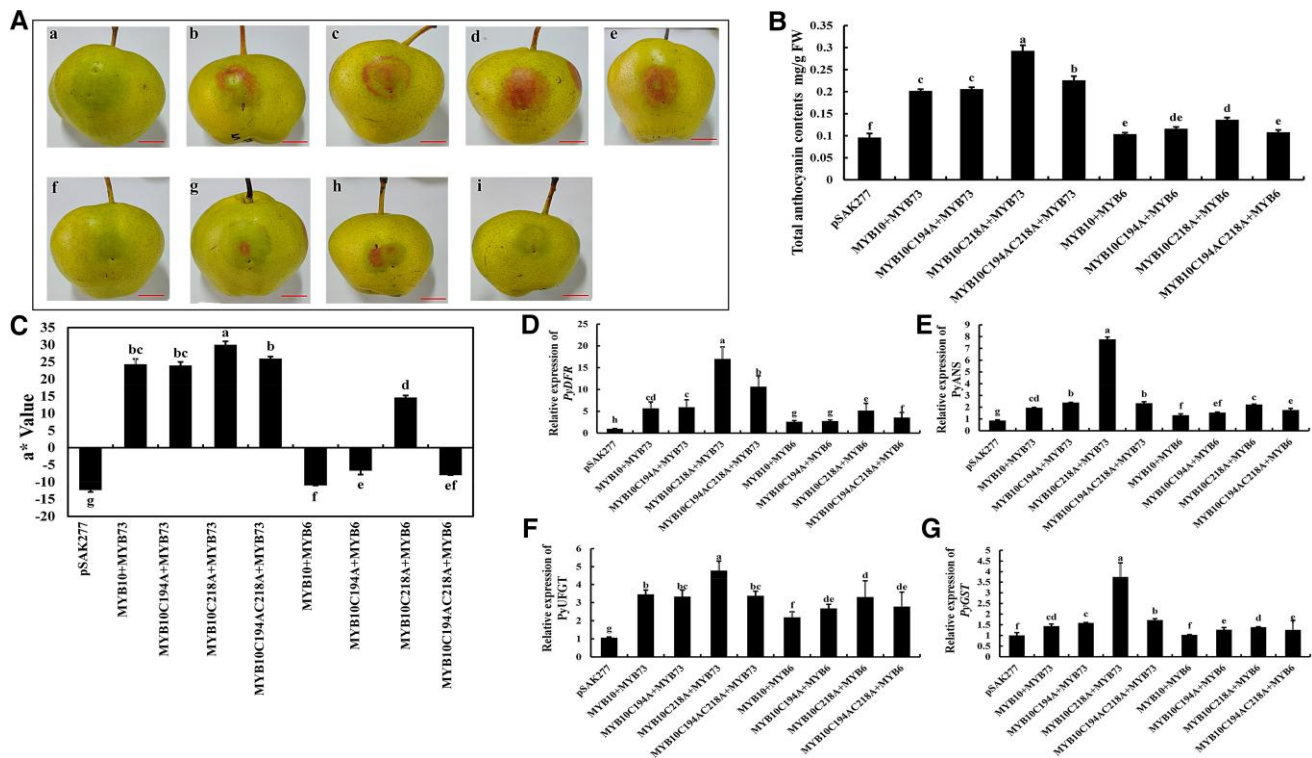


Figure 8. Cotransformation of the persulfidation modification of PyMYB10 and its partners resulted in anthocyanin biosynthesis in pear skins by transient expression assays. **A)** The phenotype of the pear cultivar ‘Zaosu’ at 4 d after infiltration: a, pSAK277; b, PyMYB10 + PyMYB73; c, PyMYB10^{Cys194Ala} + PyMYB73; d, PyMYB10^{Cys218Ala} + PyMYB73; e, PyMYB10^{Cys194AlaCys218Ala} + PyMYB73; f, PyMYB10 + PyMYB6; h, PyMYB10^{Cys194Ala} + PyMYB6; i, PyMYB10^{Cys218Ala} + PyMYB6; j, PyMYB10^{Cys194AlaCys218Ala} + PyMYB6. The scale bars represent 2 cm. **B)** The total anthocyanin contents were assessed in transformed pear peels. Bars indicate mean values $\pm$ SD from three biological replicates. **C)** Chromatism analysis of the color change in the injected regions of pear peels. “a” indicated the range from red to green; Error bars represent the mean value $\pm$ SD from six biological repeats. **D–G)** The expression levels of the genes, *PyDFR*, *PyANS*, *PyUGT*, and *PyGST* were analyzed in pears by RT-qPCR. Bars indicate mean values $\pm$ SD from three biological replicates. Statistical analysis was carried out with Student’s *t*-test, and significance was marked with different letters ($P < 0.05$).

FhMYB27 protein can repress anthocyanin accumulation by interacting with the FhTT8L, which belongs to the bHLH TF family (Li et al. 2020). Here, leucine in the EAR motif of PyMYB73 is replaced by phenylalanine, generating diverse function differences with PyMYB6 (LxLxL), suggesting that a complete EAR motif in the C-terminus is essential for its activity.

MYBs act as activators or repressors coregulated anthocyanin biosynthesis

The MYB family is large and has diverse function in plants, which are the key factors in regulating secondary metabolism and responding to biotic and abiotic stress (Wu et al. 2022). MYBs not only act as transcriptional activators but also act as repressors. MYB TFs as activators usually interact with bHLH and WD40 to form a regulatory complex. However, these three proteins are not indispensable for the anthocyanin biosynthesis. Here, we found co-expression of PyMYB73 or PyMYB6 with PyMYB10/PyMYB114 + *PybHLH3* and enhance or repress anthocyanin biosynthesis in strawberry receptacles and pears, and PyMYB73, PyMYB10, and PyMYB6 could

interact with each other (Fig. 4), whereas we detected no interaction between PyMYB114/PybHLH3 and PyMYB73/PyMYB6 in yeast. In a previous study, PyMYB10 with PyMYB114 were reported to co-regulate anthocyanin accumulation in red-skinned pear by forming PyMYB10-PyMYB114 regulatory complex (Yao et al. 2017), *PpMYB140* as a repressor to inhibit anthocyanin biosynthesis in red-skinned pears (Ni et al. 2021). Our study provided evidence that the complex PyMYB10-PyMYB73-PyMYB6 for balancing the excessive accumulation of anthocyanin in the coloration of red-skinned pear. Meanwhile, we speculate that PyMYB73/PyMYB6 may interact with PyMYB114 in another way to regulate the biosynthesis of anthocyanins. It is reported that *PyWRKY26* had a stronger activation effect on the *PyMYB114* promoter to enhance the expression of anthocyanin biosynthesis and transport-related structural genes (Li et al. 2020), indicating PyMYB73/PyMYB6 probably will act on the promoter of *PyMYB114* to regulate anthocyanin biosynthesis. Furthermore, PyMYB114 can interact directly with PyMYB10 on the protein level to substantially enhance anthocyanin biosynthesis, indicating PyMYB73/PyMYB6 could probably indirectly regulate the activity of



and PyMYB10^{Cys194AlaCys218Ala} in transient transformation in pears (Figs. 6 and 7), it suggested that H₂S-mediated persulfidation modification of PyMYB10 at Cys218 is a key modification site for regulating anthocyanin biosynthesis in red-skinned pear. Further validation showed that cotransformation of PyMYB10^{Cys218Ala} with PyMYB73 in pear fruit also promotes anthocyanin accumulation by the interaction PyMYB10 with PyMYB73, but not PyMYB6 (Fig. 8), and H₂S-mediated persulfidation weakened the interaction with PyMYB73 but enhanced the interaction with PyMYB6. These results were similar to the report from Chen et al. (2020) to the persulfidation of SnRK2.6 promotes its activity and the interaction with TF ABA response element-binding factor 2 (ABF2). It suggested that protein persulfidation could alter the interaction by partially or completely compromising. Further studies are needed to determine whether persulfidation acts by causing changes in PyMYB10 protein structure and how persulfidation of PyMYB10 regulates anthocyanin accumulation in red-skinned pear in vivo. Thus, we provided a model, in which PyMYB10 undergoes persulfidation at Cys₂₁₈, enhanced the interaction with PyMYB6 and reduced the interaction with PyMYB73, resulting in the inhibition of the expression of anthocyanin biosynthesis-related genes *PyDFR*, *PyANS*, *PyUFGT*, and

PyGST, thereby inhibited anthocyanin accumulation in red-skinned pears (Fig. 10). Our results provided a molecular mechanism for H₂S-mediated anthocyanin biosynthesis in red-skinned pears.

Materials and methods

Plant materials

In this study, the red-skinned pear ‘Hongzaosu’ (*Pyrus pyrifolia* Nakai) and its green-skinned pears ‘Zaosu’ were gathered at the 60 DAFB in 2019 from an orchard of the Institute of Pomology (Shandong Province, Tai’an). The diploid strawberry (*Fragaria vesca*, called ‘Yellow Wonder’ 5AF7) was collected for the transient transformation experiment in this study, tobacco (*N. benthamiana*) with six-leaves for the dual-luciferase reporter assay and firefly luciferase complementation assay, which were planted in an artificial incubator with a 16 h photoperiod and the temperature of 21 °C at day and 17 °C at night. ‘Zaosu’ pears, strawberry receptacles, and young *N. benthamiana* leaves with six leaves were used for the infiltrating experiment, and the observations at 5 d after injection were performed. Then collected tobacco leaves,

pear peels, and strawberry receptacles, were cut into pieces, frozen in liquid nitrogen (LN₂), and refrigerated at –80 °C.

Exogenous H₂S treatment of pear fruit and tobacco leaves

‘Aohong No.1’ pears with black-red-yellow bagging from an orchard in Anhui province Feidong County, and the pears at the ripening stage with no mechanical damage and insect pests were selected, then randomly divided into four groups. Four pear fruits in each group were fumigated in a diameter 300-mm glass dryer, and 300 mL NaHS solution (acting as H₂S donor) in the dryer with the concentration of 0.0, 0.3, 0.6, and 0.9 mM, respectively. Four-group fruits were added to four different concentration NaHS solution dryers every 24 h, and the corresponding concentration of NaHS solution was replaced. Until 72 h after the first batch of pear fruits were treated, all the pear fruits were taken out and placed in the same environment of light and temperature (25 °C). The changes in the samples were photographed and sampled at 3, 6, and 9 d after irradiation. The pear peels of the treatment group were quickly removed and stored at –80 °C by quartering method and used for further experimental research.

The method of H₂S treatment of tobacco leaves with six leaves, the experimental group and the control group were fumigated in a diameter 300-mm glass dryer containing a certain amount of water or 0.5 mM NaHS solution for 1 h, then sampled and fully ground in LN₂, and the corresponding detection reagents were added for determining.

Determination of the production of endogenous H₂S

The lead sulfide method was used to determine the production capacity of endogenous H₂S in pear pericarp. The specific steps refer to Sun et al. (2021). Weighing 0.3 to 1 g of pear peel sample and were fully ground it in LN₂. Then it was dissolved in PBS (including 10 mM Cys and 10 mM PLP, pH 6.8) buffer solution, and put the dissolved samples in a 37 °C incubator for dark treatment for 1 to 3 h to observe the experimental results. Referring to Hu et al. (2012) method to analyze the changes in H₂S production capacity.

Bioinformatic analysis

The heat map was constructed with the R script, and the phylogenetic tree was constructed by the MEGA7 Program using the neighbor-joining method, bootstrap analysis (1,000 replicates). The amino acid sequence alignment analysis was performed through the DNAMAN Program.

Measurement of anthocyanins in strawberry and pear fruits

Anthocyanins were extracted based on the method (Yao et al. 2017). Weighing about 0.2 g, the pear skins of ‘Hongzaosu’ and ‘Zaosu’ or strawberry receptacles were ground into powder in LN₂ and then added 1 ml of pre-cooled 1% (v/v) methanol hydrochloride to make it

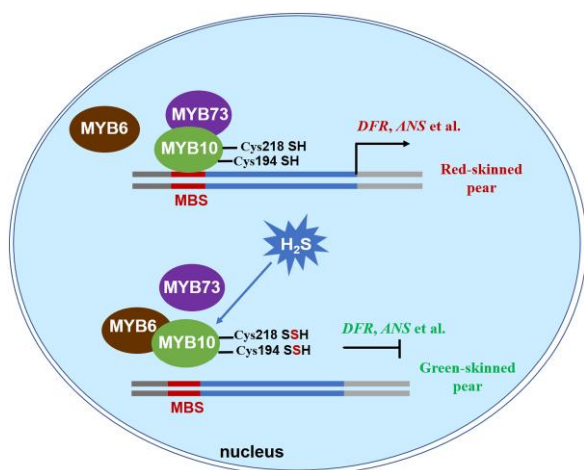


Figure 10. Proposed model of the MYB10-MYB73-MYB6 complex regulates anthocyanin synthesis when exogenous H₂S is applied or under normal conditions. In the model, PyMYB10 could interact with PyMYB73 and form a regulatory complex PyMYB10-PyMYB73, binding to the promoter of anthocyanin biosynthesis-related genes (such as *PyDFR*, *PyANS*) by MBS (MYB transaction element), resulting in enhanced anthocyanin biosynthesis in red-skinned pears. When PyMYB10 undergoes persulfidation at Cys218 and Cys194, it enhances the interaction with PyMYB6 and reduces the interaction with PyMYB73, resulting in the inhibition of the expression of anthocyanin biosynthesis-related genes *PyDFR*, *PyANS*, *PyUGT*, and *PyGST*, thereby inhibiting anthocyanin accumulation in red-skinned pears. MBS: MYB transaction element, “-Cys218 SH” indicated that the Cys218 site of PyMYB10 included a “-SH”; “-Cys194 SH” indicated that the Cys194 site of PyMYB10 included a “-SH”; “-Cys218 SSH” indicated that the Cys218 site of PyMYB10 included a “-SSH”; “-Cys194 SSH” indicated that the Cys194 site of PyMYB10 included a “-SSH.”

homogenate and culture in the dark at 4 °C for 24 h. Then, the supernatant was gathered by 12,000 rpm centrifugation for 20 min. The absorbance of anthocyanins at 530, 620, and 650 nm wavelengths was measured. There were five biological replicates in each specimen, and the total anthocyanin content per specimen was counted based on the following formula: $OD = (A_{530} - A_{620}) - 0.1 \times (A_{650} - A_{620})$.

RT-qPCR analysis

Pear skins and strawberry receptacles were ground into fine powder in LN_2 . FastPure Universal Plant Total RNA Isolation Kit (Vazyme, China) was used to extract the total RNA of the specimens. Evo M-MLV RT Premix (AG, China) was used to synthesize first-strand cDNA. RT-qPCR was conducted using SYBR® Green Premix Pro Taq HS (AG, China) in a 10 µl volume. The amplification was controlled on a LightCycler® 96 (Roche) and the raw data were analyzed with LightCycler 96.exe (Roche). The strawberry genes *FvActin* and *Fv26S* and pear *PyActin* and *PyTubulin* were used as housekeeper genes. The relative expression level of genes was counted by the $2^{-\Delta\Delta C_t}$ method. Three biological replicates were performed for all analyses and error bars. RT-qPCR's primer sequences are listed in Supplemental Table S2.

Genes isolating and recombination vector construction

The CDS sequences of *PyMYB73*, *PyMYB6*, and *PyMYB10* mutation from Cys to Ala, named as *PyMYB10^{Cys194Ala}*, *PyMYB10^{Cys218Ala}*, or *PyMYB10^{Cys194AlaCys218Ala}*, which all cloned from the cDNA of 'Hongzaosu' pear were used in the following study. The fragment amplification was carried out by using Phanta® Super-Fidelity DNA Polymerase (P501-d2, Vazyme, China), and the primer sequences for vector construction are listed in Supplemental Table S3. The *PyMYB73* and *PyMYB6* genes were integrated into the pSAK277 vector. The recombinant constructs were transformed into the *Agrobacterium* strain GV3101. In addition, we used the transient expression vector of *PyMYB10* mutation was constructed as a template, and used it to construct the vector needed for further experiments.

Dual luciferase reporter (DLR) assay

Selecting a 2.0 kb sequence upstream of *PyDFR*, *PyANS*, and *PyUGT* promoters, then isolated and integrated into the pGreen II 0800-LUC vector. The recombinant construct was transferred into *Agrobacterium* strain GV3101 containing pSoup helper plasmid. The TFs were blended with the promoters (the ratio 1:9, v/v) and then transformed into *N. benthamiana* leaves for transient expression analysis. The ratio of firefly luciferase (LUC) and renilla luciferase (REN) was detected by using the Dual-Luciferase® Reporter Assay System with the manufacturer's instructions (E1910, Promega, USA).

Yeast two-hybrid assay

To explore the interaction of different TFs by Y2H assay, *PyMYB73* is divided into five amino acid residue fragments: *MYB73^{1–100}*, *MYB73^{1–170}*, *MYB73^{1–239}*, *MYB73^{170–239}*, and *MYB73^{100–239}*. *PyMYB6* is divided into seven parts: *MYB6^{1–61}*, *MYB6^{1–117}*, *MYB6^{1–225}*, *MYB6^{1–324}*, *MYB6^{225–324}*, *MYB6^{117–324}*, and *MYB6^{61–324}*. The above fragments were ligated to the pGBKT7 vector, and the CDS fragments of candidate factors *PyMYB114*, *PybHLH3*, *PyMYB73*, *PyMYB10*, and *PyMYB10* mutation were ligated to the pGADT7 linearized vector. The vectors connected to pGBKT7 and pGADT7 were co-transformed into Y2H for interaction screening. The medium of SD/-Leu/-Trp and SD/-Leu/-Trp/-His/-Ade were used to test the transformants and interactions, respectively. The co-transformation of pGBKT7-Lam + pGADT7-T + was the negative control and pGBKT7-53 + pGADT7-T was the positive control. The primer sequences were listed in Supplemental Table S4.

Firefly luciferase complementation (FLC) assay

According to the method of Chen et al. (2008) to carry out an FLC assay of *PyMYB73* and *PyMYB6* (with no stop codon), they were cloned and inserted in pCambia1300-NLuc, and the CDS of *PyMYB73*, *PyMYB10*, and *PyMYB10* mutation were cloned and jointed with the pCambia1300-CLuc vector. The primer sequences were listed in Supplemental Table S3. The mixture of NLuc and CLuc was produced at 1:1(v:v) and infiltrated into the promoters (the ratio 1:9, v/v) and then transformed into *N. benthamiana* leaves. After three days of light, the firefly luciferase activity was measured by a stable Glo® luciferase analysis system (E2510, Promega, USA).

Isolation and purification of the integrated protein

The *PyMYB10*-MBP-tagged, *PyMYB73*-HIS-tagged, and *PyMYB6*-GST-tagged proteins were transformed in Rosetta (DE3) chemically competent cells. A certain amount of isopropyl-β-D-thiogalactopyranoside (0.1 mM) was appended, and the bacteria optical density was cultured to OD₆₀₀ 0.6 to 0.8, and then a part of the solution was added into 100 ml LB liquid medium for 16 h at 16 °C. Then the bacteria solution was collected with a centrifuge at 8,000 rpm for 30 min, the precipitate was suspended in cell lysate buffer, which contained 1 M Tris-HCl, 5 M NaOH, and 0.5 M EDTA, the protein was fragmented with an ultrasonic crusher, the mixture was centrifuged at 8,000 rpm for 30 min, and the extract was used for purification. The *PyMYB10*-MBP-tagged, *PyMYB73*-HIS-tagged, and *PyMYB6*-GST-tagged proteins were purified using pre-packed gravity columns Amylose Resin (NEB, E8021S), Glutathione Sepharose 4B (Citiva, 17075601) and Ni Sepharose 6 Fast Flow (GE, 17531806), respectively. The purification process of the protein was carried out according to the column specifications. Then part of the collected purified mixture was mixed with a loading buffer, which contained β-mercaptoethanol, and boiled for 10 min. 10% (v/v)

SDS-PAGE gel electrophoresis was performed and stained the gel with Coomassie Brilliant Blue.

S-thiolation signal detection of the recombinant protein by biotin-switch assay

Biotin-switch assay for detecting S-thiolation signal. The purified protein sample was treated with 0 to 1000 μ M NaHS at 4 °C, and the treated sample was precipitated with acetone, then the free-SH in the sample was blocked by 20 mM methyl methanethiosulfonate (MMTS) and the SSH of Cys was labeled with 2 mM biotin-HDPD, the detail steps referred to [Chen et al. \(2021\)](#). Then, the samples were subjected to 10% (v/v) SDS-PAGE gel electrophoresis. Then the proteins were transferred to 0.45 μ m polyvinylidene fluoride membrane (PVDF) and subjected to western blotting. ImageJ was used to quantify protein abundance.

Measurement of the persulfidation sites of PyMYB10 by LC-MS/MS

The persulfidation sites of PyMYB10 were conducted by LC-MS/MS, the recombinant protein was separated with nonreduced SDS-PAGE, and the target band was cut from the Coomassie brilliant blue stained gel, and then placed in a 96-well plate. Corresponding reagents are added to the 96-well plate to alkylate and digest the gel. Then it was dried by vacuum centrifugation, and the dried sample was resuspended with a solution containing 0.1% (v/v) formic acid and 2% (v/v) acetonitrile. And the sample was analyzed by LC-MS/MS (Fusion Lumos, Thermo Fisher, USA), the detailed steps referred to by [Chen et al. \(2021\)](#).

Accession numbers

Sequence data from this article can be found from the National Center for Biotechnology Information (NCBI, <https://www.ncbi.nlm.nih.gov/>), the genome database for the Rosaceae (<https://strawberry-garden.kazusa.or.jp/>), the genome database for Arabidopsis (<https://www.arabidopsis.org/>) or Solanaceae (<https://solgenomics.net/>) under the following accession numbers: PyMYB73 (Pbr008748.1), PyMYB10 (Pbr016663.1), PyMYB6 (Pbr025199.1), PyERF4.1 (ON652752), PyERF4.2 (ON652753), PcMYB10 (JX403957), PyActin (JN684184), PyTubulin (XM_009376045.2), PyERF3 (ASY06613.1), PyMYB114 (ASY06612.1), PybHLH3 (XP_048442703.1), PyANS (Pbr001543.2), PyDFR (Pbr020145.1), PyUFGT (Pbr039986.1), PyGST (Pbr012649.1), FvActin (gene22626), Fv26S (gene11892), FvANS (gene32347), FvDFR (gene15174), FvUFGT (gene12591), FvGST (gene31672) AtMYB44 (AT5G67300), AtMYB73 (AT4G37260), AtMYB70 (AT2G23290), AtMYB113 (AT1G66370) MdMYB110a (AFC88038.1), FaMYB44.1 (XP_004287994.1), FaMYB44.2 (XP_004291065.1), FaMYB44.3 (XP_004305954.1), MdMYB1 (ABK58136), MdMYBA (AB279598.1), SlMYB75 (Solyc10G086250), OsMYB44 (XP_015612590.1), StMYB44(XP_006367421).

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Author contributions

G.Y., S.G., T.Z., and H.Z. conceived this project and designed the research. G.Y., S.G., and T.Z. performed the experiments. S.G., T.Z., K.H., X.A., H.S., and C.S. analyzed the data; S.W. provided the experimental material; G.Y., S.G., and T.Z. wrote the paper; K.H., S.G., T.Z. interpreted the data, G.Y. and H.Z. revised the manuscript. All authors have read and approved the manuscript.

Supplemental data

The following materials are available in the online version of this article.

Supplemental Table S1. List of differently expressed genes (DEGs) between ‘Hongzaosu’ and ‘Zaosu’ pear fruits at 60 DAFB.

Supplemental Table S2. RT-qPCR primers were used in this study.

Supplemental Table S3. The list of primers used in this study for transient expression and prokaryotic expression.

Supplemental Table S4. The list of primers used in this study for Y2H.

Supplemental Figure S1. Production of endogenous H₂S during the pear fruit development.

Supplemental Figure S2. Heatmap was used to analyze the expression levels of genes related to anthocyanin metabolism and regulation in ‘Hongzaosu’ and ‘Zaosu’ pears at 30, 60, and 90 DAFB, respectively.

Supplemental Figure S3. Coinjection of PyMYB73/PyMYB6 by transient expression assay in pear skins.

Supplemental Figure S4. Verifying the regulatory pattern of PyMYB73/PyMYB6, PyMYB10 in vivo.

Supplemental Figure S5. PyMYB73/PyMYB6 persulfidation-site and persulfidated protein assays.

Supplemental Figure S6. Dual luciferase reporter (DLR) assays verify that PyMYB10 with the mutated thiolated modification site activates the PyANS, PyDFR, and PyUFGT promoters.

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Conflict of interest statement. The authors declare that they have no conflict of interest.

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